The Fourier phase sphere: a method for computer-assisted atonal composition

Samuel Pereira | Gilberto Bernardes



2020

Agenda

I. Context

II. Preliminary concepts

III. Fourier phase sphere

... Interlude ...

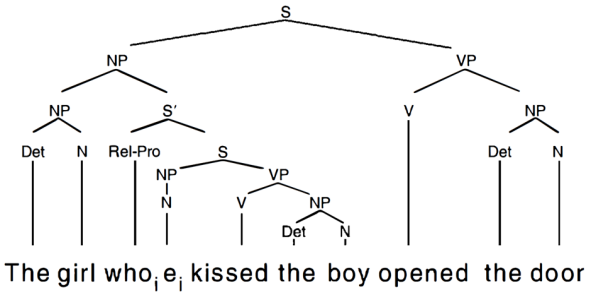
IV. Case Study

V. Final considerations

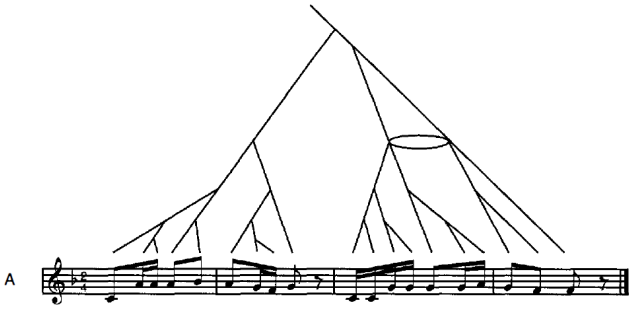
I. Context

a) Problem hypotheses

- Harmonic lexicon and syntax in (a)tonal music;



(Adapted from Patel [2010])



(Adapted from Large, Palmėr and Pollack [2005])

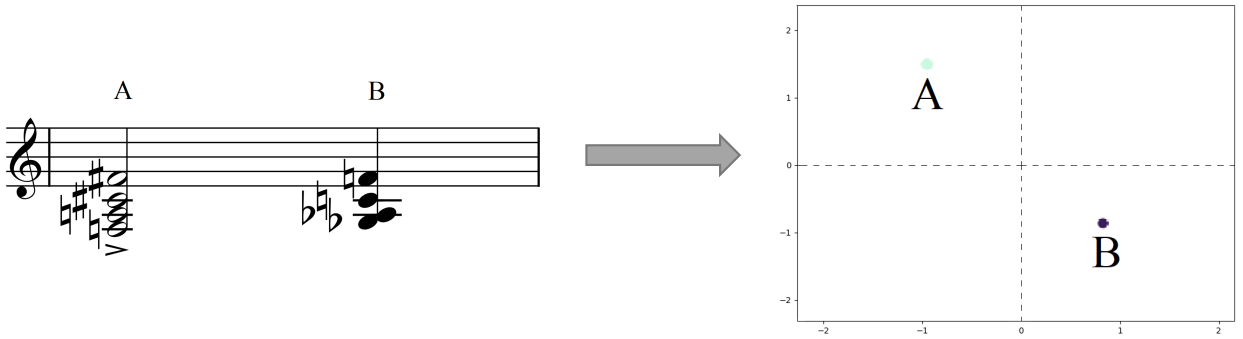
b) Objectives:

- Computationally we must...:
  - > Plot pitch-class sets (pc sets) in a geometric space;
  - > Group such pc sets into clusters with distinct characteristics;
  - > Identify trajectories in space that infer a syntactic organization.

- ... in order to assist contemporary composers in their harmonic decision-making process.

II. Preliminary Concepts

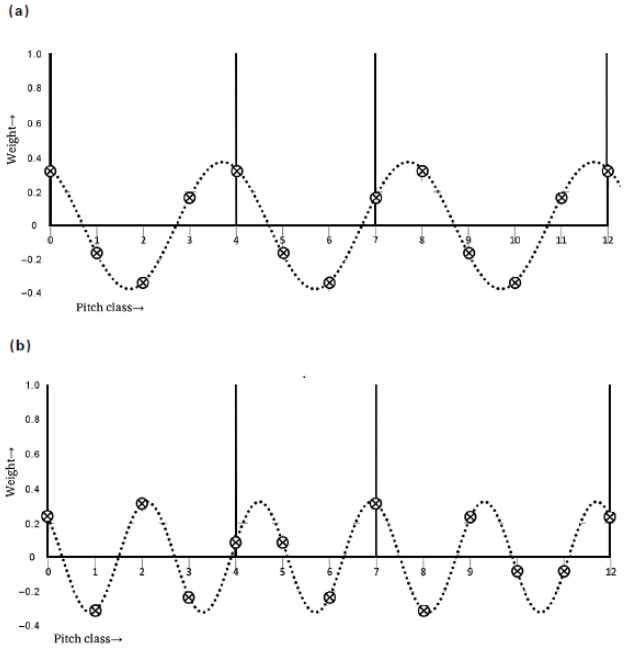
a) Geometric spaces of harmonic representation:



b) The discrete Fourier transform (DFT) of pc sets:

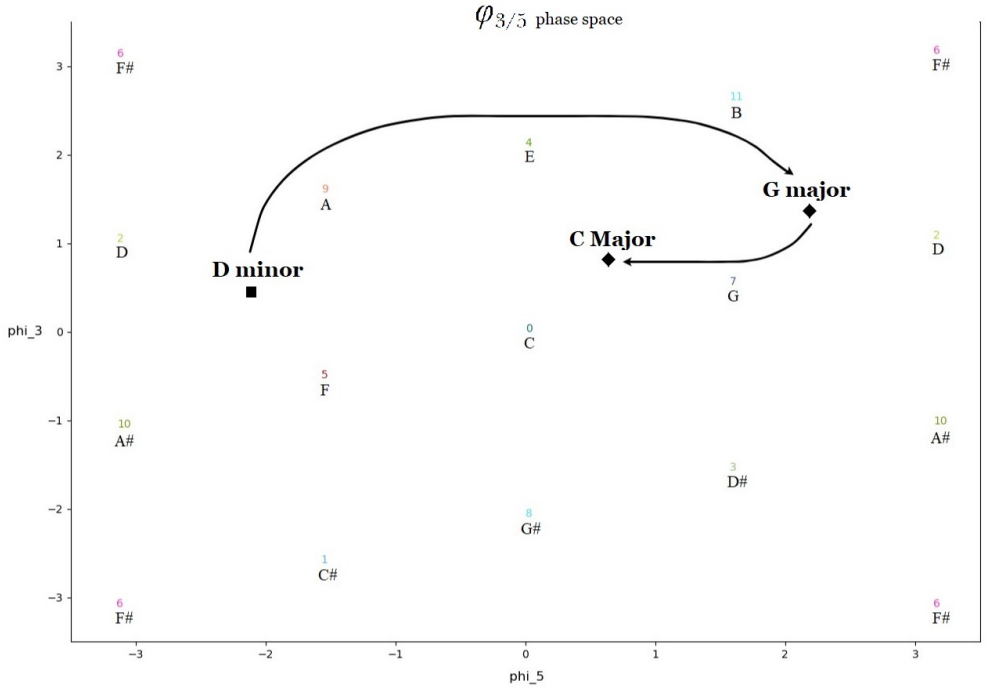
- Twelve components ($\hat{f}_n$, i.e., $\hat{f}_{0-11}$). Each one has magnitude ($|\hat{f}_n|$) and phase (φ_n) values;

- It is possible to create a two dimensional geometric space of harmonic representation using phase values of two DFT components ($\boldsymbol{\varphi_{3/5}}$).



(Adapted from Yust [2015])

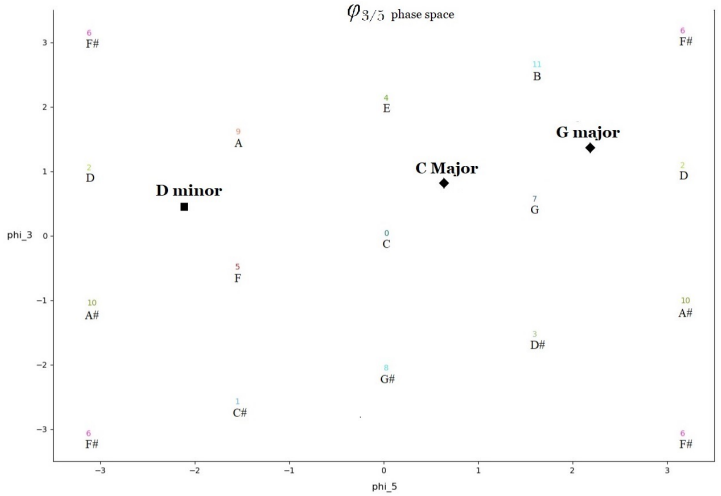
c) Yust's phase space ($\boldsymbol{\varphi_{3/5}}$):

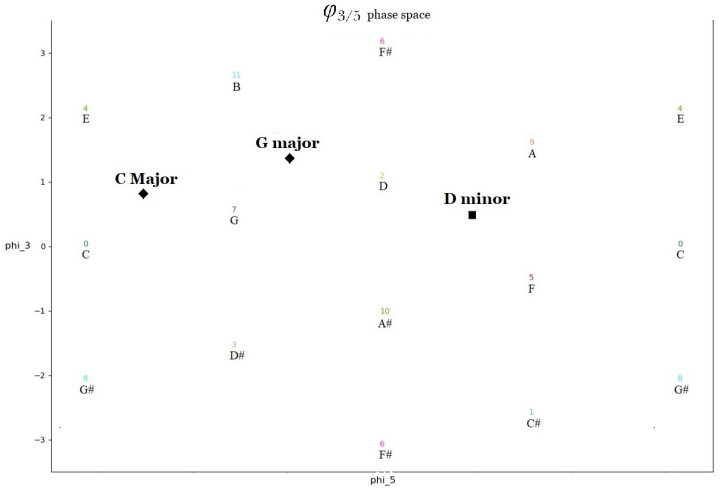


d) Obstacles:

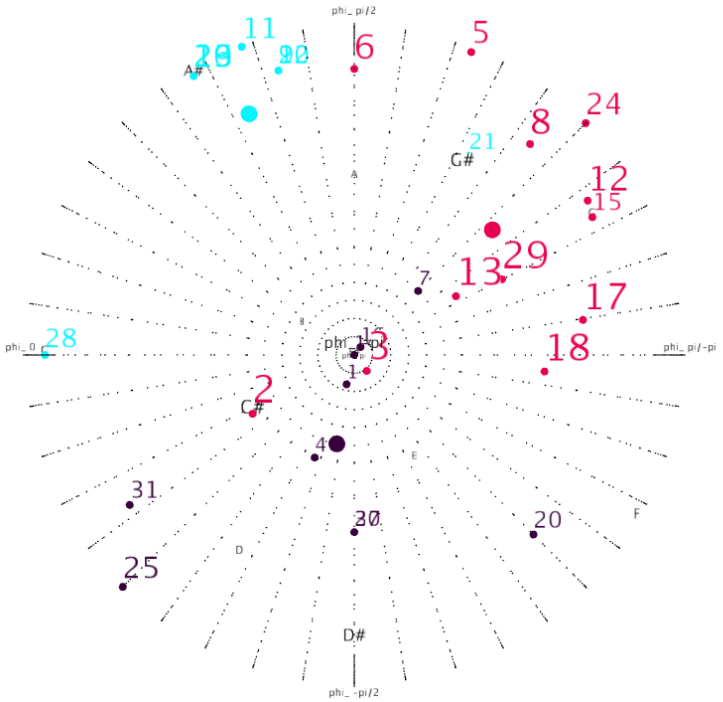
- Cyclical character in DFT phase values;

- The definition of the phase space ($\varphi_{?/?}$) to be used in the context of atonal music;





III. Fourier phase sphere



spherical phase space

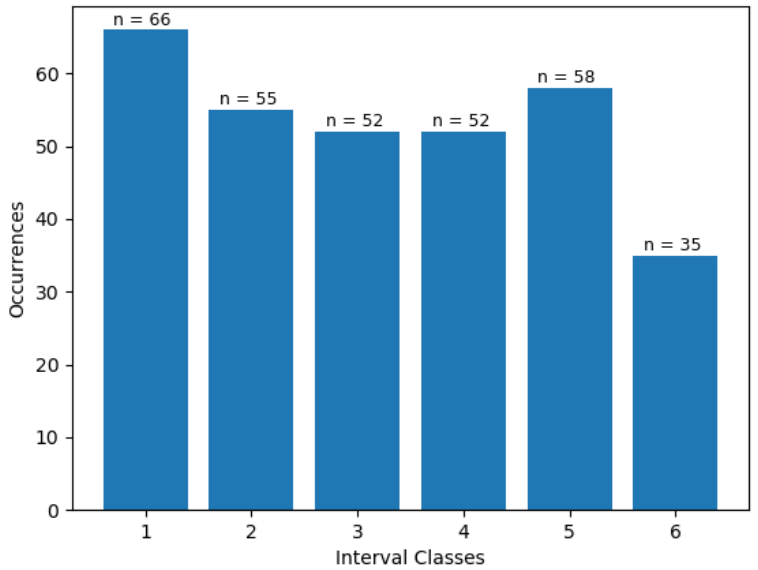
phi 1/5

v_0.2

a) What phase space($\varphi_{?/?}$)?

- Definition by counting interval classes;

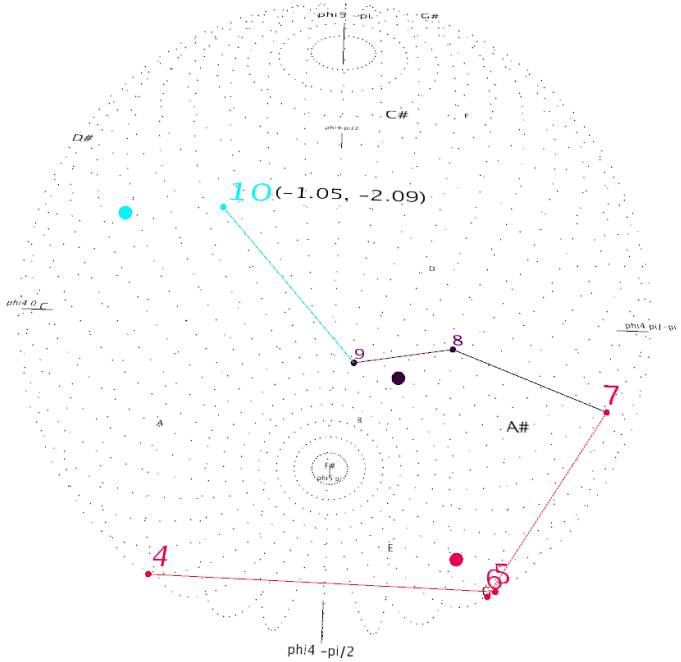
-In this example phase space $\varphi_{1/5}$ is selected.



b) Why a sphere?

- It allows the plotting of any pc set with latitude and longitude concepts;

- Concedes circularity to phase values with a simpler topology than a torus.



c) Technical details:

- Python scripts and Processing sketches;



- "Salami slicing" using *music21* library from a .xml file.

c) Technical details:

- Counts interval classes and saves two most proeminent;

- Transforms pc sets in a 12-dimensional vector where binary activation denotes the presence of one pc.

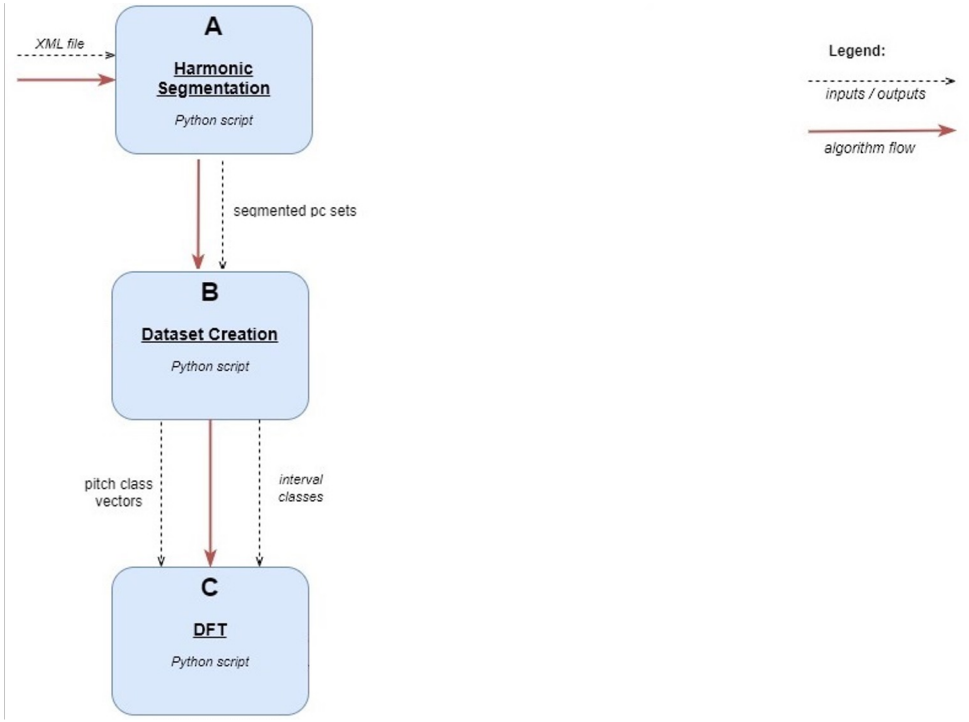


[C, E, G] ⟶ [0, 4, 7] ⟶ [1, 0, 0, 0, 1, 0, 0, 1, 0, 0, 0, 0]

c) Technical details:

- Calculates the DFT for each pc set;

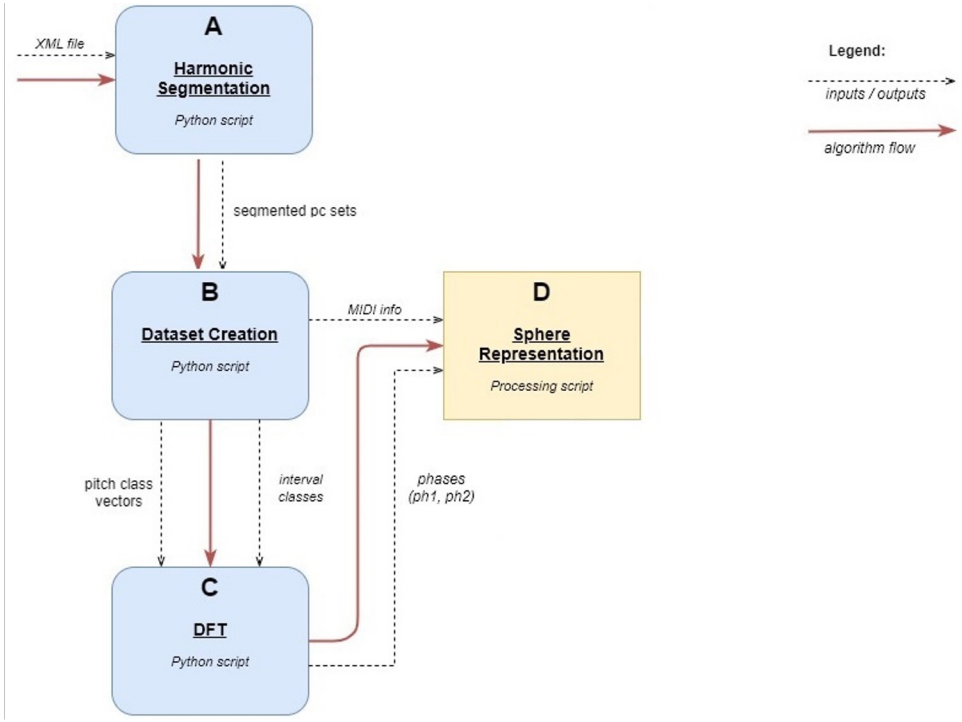
- Communicates with Processing and sends the two required phase values.



c) Technical details:

- Draws the sphere;

- Calculates 3 cartesian coordinates given two phase values in radians for each pc set.

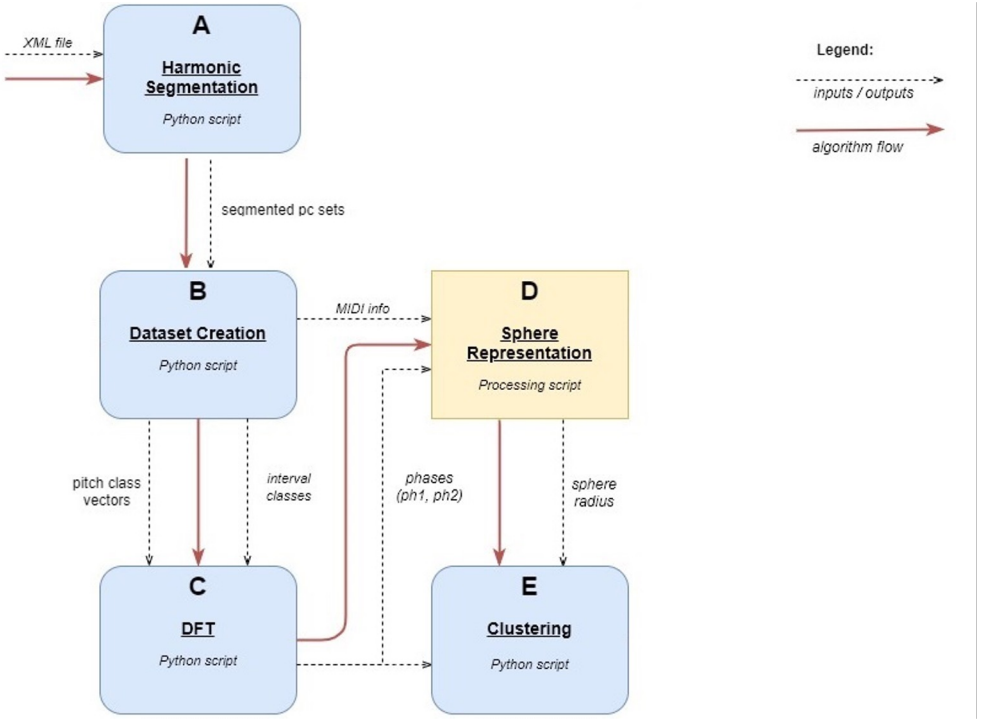


c) Technical details:

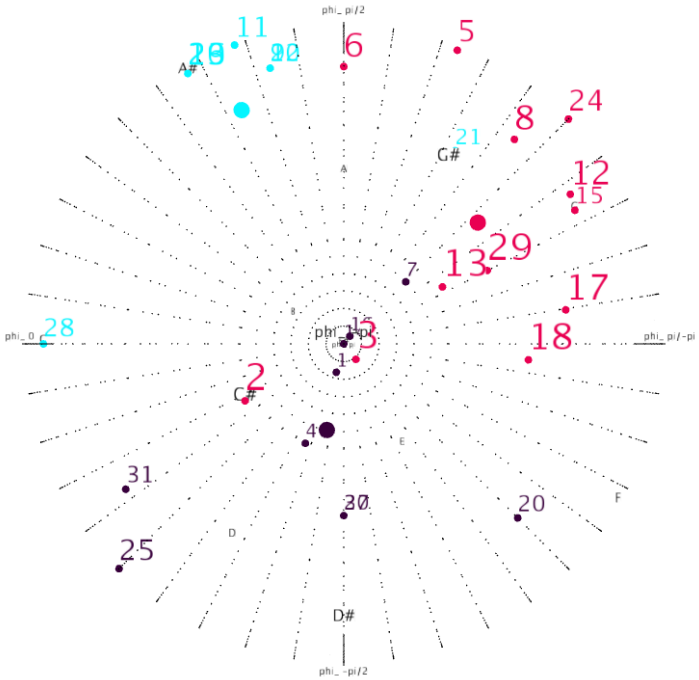
- Adopts *k-means* algorithm;

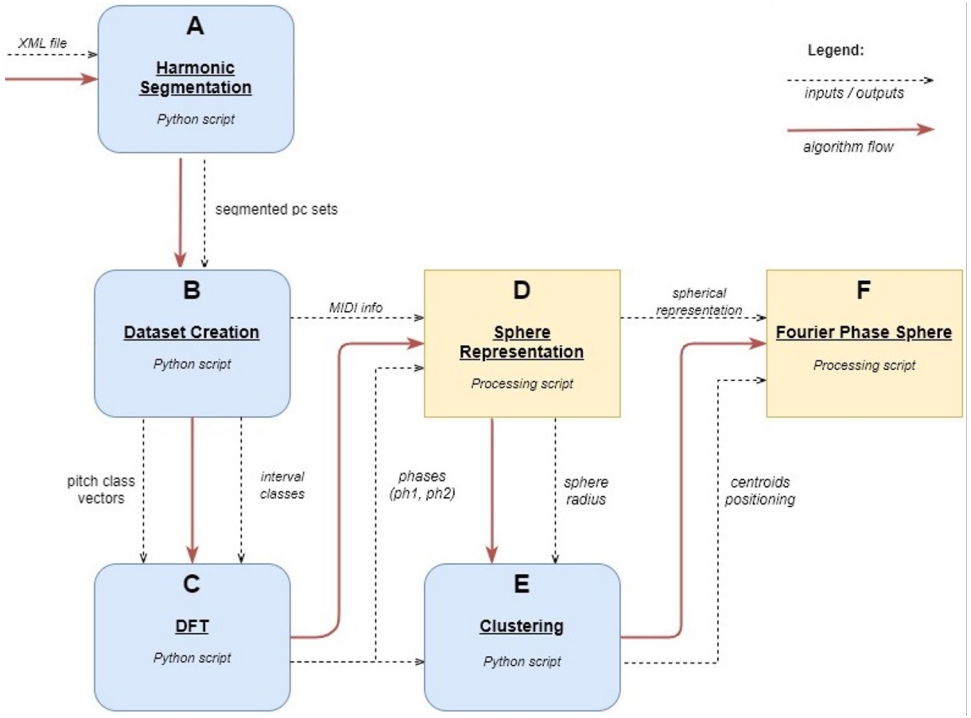
- Defines the optimal number of clusters with the "elbow method";

- Communicates to Processing the cluster to which each pc set belongs.



c) Technical details:





...interlude...

So far we...:

- ...have developed a DFT based spherical space to plot pc sets and compare its positioning;

...and...

- ...have been able to qualify harmonic information with clustering.

Can one infer an organization of harmonic syntax through geometric trajectories in space?

IV. Case study

- String quintet for two violins, two violas and cello written with Fourier phase sphere;

- Six small sections in forty five measures;



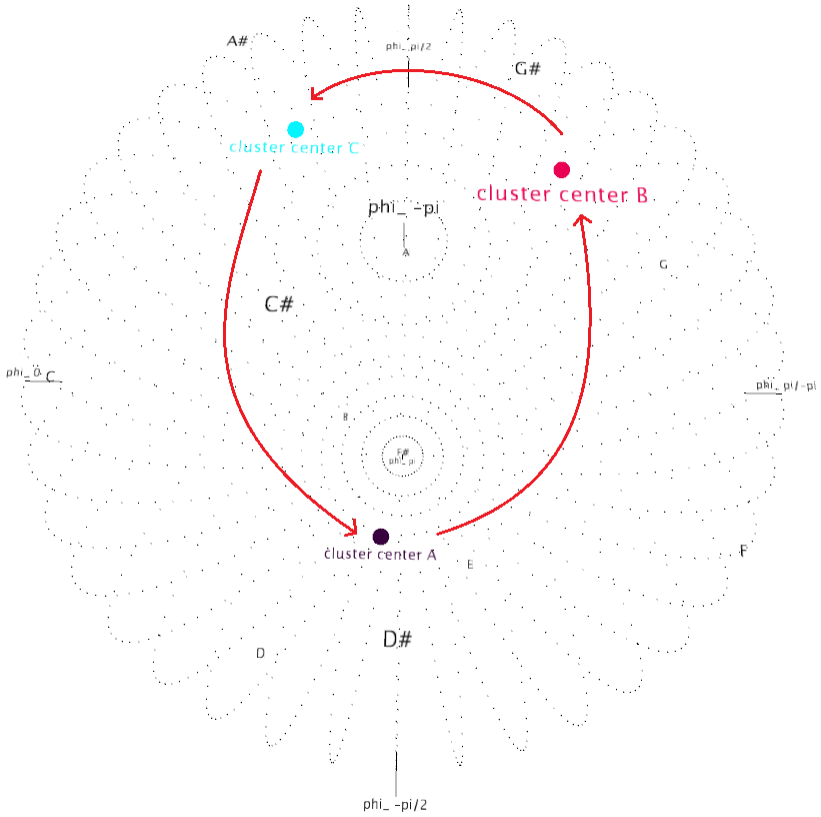
a) Compositional method:

- Lexicon definition;



a) Compositional method:

- Geometrical trajectory chosen - syntactic element between pc sets;



- Example:


I.
Quinteto para cordas
Escrito com recurso à EFF
Junho 2020
2
19
20
Violino I
Violino II
Viola I
Viola II
Violoncelo
s. t.
c. sor.
senza vib.
s.p.
s.t.
al niente
nat.
s. sor.

V. Final considerations

- Main contributions;

- Future work.

References:

AMIOT, Emmanuel. The Torii of Phases. *Mathematics and Computation in Music Lecture Notes in Computer Science,* Berlin, vol 7937, (2012-3), pp. 1-18.

AMIOT, Emmanuel. *Music Through Fourier Space:* discrete Fourier transform in music theory. Springer, 2018.

BERNARDES, Gilberto; COCHARRO, Diogo; CAETANO, Marcelo; GUEDES, Carlos & DAVIES, Matthew. A multi-level tonal interval space for modelling pitch relatedness and musical consonance. *Journal of New Music Research*, *45*(4), 2016, pp. 281-294.

FERGUSON, Sean; PARNCUTT, Richard. Composing in the flesh: perceptually informed harmonic syntax. *Proceedings of Sound and Music Computing,* 2004.

LARGE, E. W., PALMÉR, C., & POLLACK, J. B. Reduced Memory Representations for Music. *Cognitive Science*, 19(1), 1995, pp. 53-96.

LEWIN, David. Re: Intervallic Relations between Two Collections of Notes. *Journal of Music Theory*, *3*(2), 1959, p. 298.

LEWIN, David. Special Cases of the Interval Function between Pitch-Class Sets X and Y. *Journal of Music Theory*, *45*(1), 2001, p. 1.

PATEL, A. D. *Music, language, and the brain*. Oxford: Oxford University Press, 2010.

PEREIRA, Samuel. *A Sintaxe Harmónica Atonal no Espaço de Fourier*: Para Uma Representação Esférica das Fases. Porto, 2020. 145p. Dissertação do Mestrado em Multimédia da Universidade do Porto.

QUINN, Ian. General Equal-Tempered Harmony (in two parts). *Perspectives of new music,* 44/2; 45/1, 2006-7, pp. 114-59; pp. 4-63.

ROHRMEIER, Martin; PEARCE, Marcus. Musical Syntax I: Theoretical Perspectives. *Springer Handbook of Systematic Musicology Springer Handbooks,* 2018, pp. 473-486.

THORNDIKE, Robert. Who Belong in the Family? *Psychometrika,* 1953, 18 (4), pp. 267-276.

YUST, Jason. Schuberts Harmonic Language and Fourier Phase Space. *Journal of Music Theory*, 59(1), 2015, pp. 121-181.

YUST, Jason. Special Collections - Renewing Set Theory. *Journal of Music Theory*, 60(2), 2016, pp. 213-262.

YUST, Jason. Generalized Tonnetze and Zeitnetze, and the topology of music concepts. *Journal of Mathematics and Music,* 2020, pp. 1-35.

Thank you!